Beyond Translation: Evaluating Logical Reasoning Competence in Large Language Models under Turkish Morphosyntactic Variations

A Reproducible Framework, Construction Audit, and Evaluation Protocol

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Abstract

Evaluating an answer in a single language and formulation provides an incomplete account of logical reasoning reliability. For Turkish, case marking, possessive constructions, phonologically conditioned suffixes, and flexible constituent order introduce meaningful evaluation dimensions that sentence translation alone does not control. We present Turkish-CBE, a reproducible framework for comparing answers to the same latent ordering problem across Turkish and English and across answer-preserving Turkish presentations. The framework separates an exact symbolic generator from linguistic realization, distinguishes endpoint from internal and distance-controlled queries, and evaluates both correctness and stability on explicitly declared denominators. The frozen design contains 3,000 latent orders, 23,200 source queries, and 5,637,600 derived prompt identities. We report construction statistics and independently measured lexical/tokenization features, rather than unperformed model experiments. All 18 canonical Turkish relation expressions have zero exact surface counts in the retained corpus measurement, despite positive lemma counts, illustrating the sensitivity of familiarity proxies to morphological realization. Tokenizer measurements expose variation across languages and relation directions; these are descriptive preprocessing observations, not evidence of a reasoning deficit. We formulate a paired statistical protocol, show how valid-pair coverage bounds the interpretation of robustness, and distinguish the pooled-marginal CBE chance correction from conventional Cohen agreement. Human linguistic adjudication and genuine model evaluation remain prerequisites for answering the empirical research questions.

1 Introduction

A logical conclusion can remain unchanged when its premises are reordered, a relation is expressed through its converse, or a language uses a different grammatical realization. A model that changes its answer under such transformations presents a different evaluation problem from a model that consistently reaches the wrong conclusion. Aggregate accuracy obscures both distinctions: equal scores can conceal failures on different instances, while high agreement can arise from repeating a biased answer. Behavioral tests and consistency evaluations therefore complement conventional task-level scores (Ribeiro et al., 2020; Elazar et al., 2021).

The problem is particularly consequential for multilingual evaluation. Turkish expresses grammatical relations through productive suffixation and morphophonological alternations, and permits constituent-order choices whose discourse effects are not captured by a literal translation. Computational descriptions of Turkish morphology formalize the interaction between morphotactics and phonological realization (Oflazer, 1994). Recent Turkish evaluations investigate morphological compositionality, linguistic minimal pairs, and surface-form confounders (Ismayilzada et al., 2025; Başar et al., 2025; Başar and Bisazza, 2026). These studies motivate controlled linguistic testing, but they do not establish how equivalent Turkish formulations affect transitive ordering inference. That question requires its own matched design and genuine model outcomes.

Competence-Based Evaluation (CBE) treats correctness across presentations of a parameterized problem family as a behavioral requirement (Jiang and Zhai, 2026). We adopt this framing and extend it to

Turkish–English matching, morphosyntactic interventions, lexical measurements, and query-distance controls. The object of comparison is the same latent instance and question, rather than unrelated language-specific test items. A failure on one equivalent formulation falsifies the corresponding finite-instance requirement; success on a finite sample does not certify an unbounded reasoning competence or reveal its internal implementation.

Turkish-CBE addresses five research questions. RQ1 concerns the paired Turkish–English accuracy difference. RQ2 concerns sensitivity to answer-preserving Turkish presentations. RQ3 separates entity count from query distance and query type. RQ4 examines associations among relation vocabulary, independent corpus frequency, tokenization, and predicted-answer bias. RQ5 asks whether reasoning prompting or optional adaptation improves correct consistency as well as accuracy. We specify estimands and controls for these questions without presupposing that Turkish is harder, that morphology causes an error, or that an intervention necessarily helps.

The present article has three contributions. First, it describes a deterministic, language-independent ordering generator with exact answers and split-level separation of latent instances. Second, it specifies Turkish rendering and bilingual matching with explicit limits on what automatic semantic checks can establish. Third, it provides a denominator-aware reliability analysis and reproducible visual audit of the constructed data and measured lexical features. This is a framework and construction/preprocessing study: real-model performance is not yet available, and the manuscript makes no empirical claim of a language gap, prompting gain, or training benefit.

2 Related Work

2.1 Logical reasoning benchmarks and controlled generation

Natural-language reasoning benchmarks differ in their source material and the inferential structures they expose. ReClor and LogiQA evaluate reasoning-oriented reading comprehension (Yu et al., 2020; Liu et al., 2020). FOLIO connects natural-language arguments to first-order logical representations (Han et al., 2024). Their task diversity is valuable, but a change in the language or surface form of an item need not preserve all other evaluation conditions. Turkish-CBE instead fixes a narrow ordering family so that the latent question and its answer can be held constant across presentations.

Generated benchmarks provide complementary structural control. CLUTRR tests relational reasoning from text (Sinha et al., 2019); RuleTaker studies reasoning over natural-language rule bases (Clark et al., 2020); and ProofWriter adds generated implications, proofs, and abductive statements (Tafjord et al., 2021). PrOntoQA supports formal analysis of reasoning chains (Saparov and He, 2023). These precedents show why explicit structure and an oracle are useful. Our contribution is not the invention of generated reasoning data; it is their use within a matched Turkish presentation design, with input size and latent proof distance separately recorded.

2.2 Robustness, metamorphic relations, and consistency

Metamorphic testing specifies relations between outputs under controlled input transformations (Segura et al., 2016). CheckList applies behavioral test design to language technologies (Ribeiro et al., 2020). ParaRel studies consistency across paraphrased factual queries (Elazar et al., 2021); factual cloze consistency is distinct from deductive ordering, but it motivates separating correctness from stability. CBE combines a scalable latent family with an explicit answer-preservation requirement (Jiang and Zhai, 2026). We retain that requirement and add Turkish-specific interventions and missing-response accounting.

Controlled perturbations in arithmetic also motivate careful interpretation. GSM-Symbolic evaluates variation in mathematically structured problems (Mirzadeh et al., 2025). Its task differs from ordering, so neither its numerical findings nor a particular mechanistic explanation is imported here. A presentation effect can reflect parsing, retrieval, lexical priors, or inference; behavioral inconsistency alone does not

distinguish those causes.

2.3 Multilingual reasoning and Turkish linguistic evaluation

MGSM investigates multilingual chain-of-thought arithmetic reasoning (Shi et al., 2023). Turkish is not one of its evaluated languages, and arithmetic success cannot validate Turkish ordering templates. A multilingual ordering benchmark must therefore establish its own linguistic equivalence and matched sampling.

Turkish morphology has long motivated explicit computational modeling (Oflazer, 1994). Morphological compositionality experiments expose limitations in combining morphemes productively (Ismayilzada et al., 2025). TurBLiMP evaluates Turkish minimal-pair acceptability (Başar et al., 2025), whereas morphology-aware syntax evaluation examines how surface factors can bias judgments (Başar and Bisazza, 2026). Those tasks are complementary to ours: minimal-pair grammaticality contrasts often intentionally change acceptability, while Turkish-CBE requires grammatical alternatives that preserve truth and the gold answer. Native-speaker adjudication remains necessary despite deterministic rendering.

2.4 Tokenization, frequency proxies, and reasoning interventions

Tokenizer quality and language coverage can affect multilingual model behavior (Rust et al., 2021). Token-length inequalities also matter for access costs and processing budgets (Ahia et al., 2023; Petrov et al., 2023). These observations justify measuring tokenization rather than assuming equal cost for matched text. They do not establish a causal tokenization explanation for a logical error. Corpus frequency is likewise a proxy with explicit genre, annotation, and segmentation limits; our independent resource is the locked BOUN treebank release (Türk et al., 2022).

Chain-of-thought prompting and self-consistency are distinct interventions (Wei et al., 2022; Wang et al., 2023). Self-consistency aggregates sampled responses to one input; presentation invariance compares different inputs that preserve the same answer. Moreover, generated explanations need not faithfully report the causes of an answer (Turpin et al., 2023). Parameter-efficient adaptation with LoRA (Hu et al., 2021) and group-relative policy optimization (Shao et al., 2024) motivate optional training arms, but their success elsewhere provides no evidence that our unrun adapters improve Turkish reasoning.

3 Methodology

3.1 Latent structure, oracle, and query geometry

Let a latent instance be a strict total order

$$z = (e_{\sigma(1)} \prec e_{\sigma(2)} \prec \cdots \prec e_{\sigma(n)}), \qquad |E| = n. \tag{1}$$

The frozen size grid is $n \in \{4, 8, 16, 32, 64, 128, 256, 512, 1024\}$. The generator draws a permutation of unique entities and emits the $n - 1$ adjacent comparisons. A language-independent rank map $r_z : E \to \{1, \ldots, n\}$ defines the exact answer to a forward query:

$$y^*(z, a, b, \prec) = \mathbf{1}\{r_z(a) < r_z(b)\}. \tag{2}$$

The inverse relation reverses the inequality. A separate graph check validates the chain and its entailed comparisons. Stable, versioned seed derivation separates naming, permutation, premise order, and query selection, so introducing a new presentation condition does not alter previously defined latent instances.

Endpoint queries compare the first and last positions. Internal queries sample two positions strictly inside the chain. Distance-controlled queries fix

$$d = |r_z(a) - r_z(b)|, \tag{3}$$

Table 1: Turkish presentation factors. P0 denotes the canonical form. The general matched equivalent condition is retained separately from isolated ablations.

Factor	Operation	Interpretation control
P0	Canonical grammatical realization	Baseline; identical latent query.
P1	Reorder premises and names list	Preserve words and predicates.
P2	Swap arguments and use inverse relation	Preserve the asserted comparison.
P3	Alternative constituent construction	Truth preservation does not imply identical discourse preference.
P4	Valid case/possessive morphology alternatives	Label additional syntax/lexical changes explicitly.
P5	Combine several operations	Compound perturbation; no pure morphology attribution.
P6	Optional controlled negation	Excluded from the frozen core; requires strict-order proof.

with both positions internal. Thus their feasible distances satisfy $1 \leq d \leq n-3$; the retained power-of-two grid is intersected with that range. This is narrower than merely requiring $d < n$. At $n = 4$, an internal pair necessarily has $d = 1$, so internal and distance-controlled conditions can share geometry. Such repetitions are not independent evidence.

A longer endpoint chain does not prove deeper inference. The source and sink can be identified by their incident-edge signatures under the chain assumptions, avoiding a full transitive derivation. Entity count measures input and presentation load; d measures latent graph distance, not observed model computation. We therefore retain endpoint and internal/distance strata and do not treat their pooled score as an inference-depth curve.

3.2 Language realization and equivalent presentations

A renderer $\pi_{\ell,p,v,c}$ maps a latent query to language ℓ, presentation p, relation vocabulary v, and name control c. Its formal contract is

$$\mathcal{N}(\pi_{\ell,p,v,c}(z)) = \mathcal{N}(z), \qquad y^*(\pi_{\ell,p,v,c}(z)) = y^*(z), \tag{4}$$

where $\mathcal{N}$ is the closed-grammar logical normal form. Converse realization swaps arguments and uses the inverse predicate; for example, $a \prec b$ is realized as either a forward statement about a or an inverse statement about b. Premise order and names-list order are independent surface factors.

The inventory contains nine relations from spatial, temporal, comparative, and ranking families. Spatial relations have explicitly linear positional interpretations. Turkish realization selects genitive/possessive or ablative constructions as appropriate, handles proper-name apostrophes, and applies vowel and consonant harmony. The reference to Turkish morphological theory (Oflazer, 1994) motivates these constraints; it does not certify our implementation. Table 1 separates operational transformations from linguistic claims.

3.3 Semantic validation and human adjudication

Automatic validation checks distinct entities, chain consistency, premise truth, query distance, normal-form equivalence, and oracle agreement. Rendering and parsing are partly based on shared linguistic code, creating a common-mode failure risk. Passing a round trip establishes a contract within the implemented grammar; it is not an independent proof of naturalness or unrestricted Turkish equivalence.

Two independent Turkish-proficient reviewers must judge grammaticality, naturalness, semantic equivalence, and morphology, with disagreements resolved before an approved review record is issued. The operational large-run gate requires at least 100 distinct reviewed Turkish prompts, at least 20 in each selected relation–presentation–name template cell, and coverage of every requested size and query type. This is sampled review, not exhaustive review of every cross-cell. Existing automatic checks and an unfilled

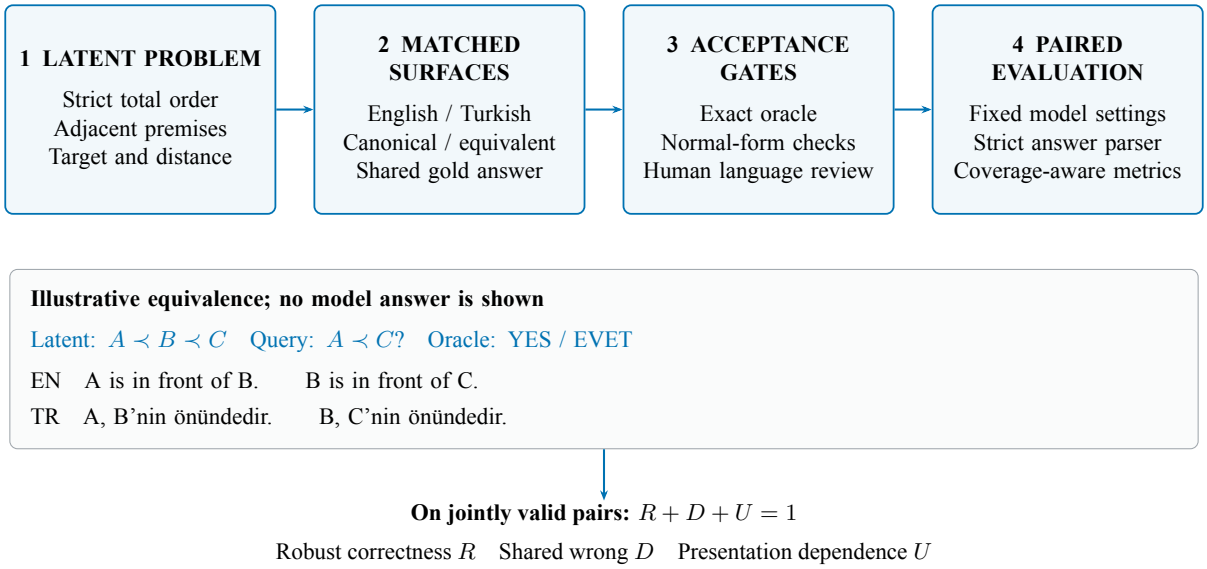


Figure 1: Turkish-CBE architecture and an illustrative bilingual realization. The diagram describes the evaluation design, not observed model accuracy.

review sheet do not constitute human approval. Work on Turkish minimal pairs reinforces the need to separate grammatical evaluation from logical labels (Başar et al., 2025; Başar and Bisazza, 2026).

3.4 Bilingual matching, names, and split integrity

Turkish and English records share the latent-order identifier, query target, query direction, gold label, and presentation plan. The `native_shared` control uses the same native Unicode spelling in both languages. `en_transliterated` changes only the English labels, retaining native Turkish spelling and morphology. `symbolic_shared` assigns identical symbolic labels in both languages. Consequently, the transliterated arm intentionally changes naming together with language; it is a naming-sensitivity control, not a pure language contrast. The shared-name arms support the cleaner matched comparison. They do not remove all differences in tokenizer vocabulary, instructions, or response labels. All derivatives of a latent order are assigned to one split before rendering; changing the vocabulary cannot move the same instance from training into test.

The upstream generator supports several entity inventories, but the frozen paired bank analyzed here uses the Turkish-name source inventory before alias expansion; a fully crossed synthetic-name experiment is not established by these counts. The unit of independence is the source order, not a prompt record or comparison cell. The total-order family is structurally isomorphic under renaming: exact train/test instance separation is not separation of abstract reasoning algorithms or a guarantee against external pretraining contamination. The current frozen grid crosses three name controls, nine relations, and nine language/presentation conditions: two English conditions and seven Turkish conditions (canonical, general equivalent, and P1–P5). This produces $3 \times 9 \times 9 = 243$ prompt identities for each source query. Optional negation is not silently included in that expansion.

4 Correctness, Invariance, and Statistical Inference

4.1 Denominators and the paired outcome decomposition

For N planned matched presentation pairs, let $a_{ik} \in \{0, 1, \perp\}$ be the strictly parsed answer to presentation $k \in \{0, 1\}$, where $\perp$ denotes invalid or missing output. Marginal correctness is $C_{ik} = \mathbf{1}\{a_{ik} = y_i^*\}$, with invalid answers wrong. Define $V_i = \mathbf{1}\{a_{i0} \neq \perp, a_{i1} \neq \perp\}$ and $M = \sum_i V_i$. The marginal accuracies and valid-pair coverage are

$$\widehat{A}_k = \frac{1}{N}\sum_{i=1}^{N} C_{ik}, \qquad \widehat{v} = \frac{M}{N}. \tag{5}$$

For $M > 0$, the paired CBE quantities use exactly the same jointly valid support:

$$\widehat{R} = \frac{1}{M}\sum_i V_i C_{i0} C_{i1}, \tag{6}$$

$$\widehat{D} = \frac{1}{M}\sum_i V_i (1 - C_{i0})(1 - C_{i1}), \tag{7}$$

$$\widehat{U} = \frac{1}{M}\sum_i V_i \mathbf{1}\{C_{i0} \neq C_{i1}\}. \tag{8}$$

Consequently,

$$\widehat{R} + \widehat{D} + \widehat{U} = 1, \quad \widehat{A}_{\text{valid}} = \widehat{R} + \widehat{U}/2, \quad \widehat{I} = \widehat{R} + \widehat{D}. \tag{9}$$

Here $\widehat{A}_{\text{valid}}$ is the mean of the two accuracies *on jointly valid pairs*. It is not automatically $(\widehat{A}_0 + \widehat{A}_1)/2$ on the full planned sample. Binary gold labels shared within each pair imply that D is shared-wrong agreement, whereas U is presentation dependence.

For $N > 0$ and $M > 0$, strict robust correctness retains all planned pairs:

$$\widehat{R}_{\text{strict}} = \frac{\sum_i V_i C_{i0} C_{i1}}{N} = \widehat{v}\,\widehat{R}. \tag{10}$$

The equality exposes selective coverage directly. It does not assert that marginal accuracy equals coverage times valid-pair accuracy: pairs with one valid, correct answer can still contribute to a marginal. Both response-level invalid rates and pair coverage must accompany every conditional profile. If $M = 0$ and $N > 0$, strict robustness is zero while conditional R is undefined; the product identity is used only when R is defined. Empty conditional denominators are undefined, not zero.

4.2 Chance correction and its interpretation

At a fixed complexity and declared model/condition stratum, define q as the pooled YES/EVET rate over the $2M$ jointly valid answers. The retained CBE reference is

$$I_e = q^2 + (1 - q)^2, \qquad \kappa_{\text{CBE}} = \frac{\widehat{I} - I_e}{1 - I_e}. \tag{11}$$

Constant outputs make $I_e = 1$, so the coefficient is undefined. High raw agreement from a constant answer is therefore not evidence of competence. The chance correction is related to the general idea of agreement beyond chance (Cohen, 1960), but its pooled reference is not identical to the separate-marginal reference in conventional Cohen kappa. Appendix A derives the difference. Neither reference identifies an internal reasoning mechanism or proves stochastic independence.

A valid-pair mean accuracy alone bounds but does not determine robustness. From the nonnegative outcome masses,

$$\max(0, 2A_{\text{valid}} - 1) \leq R \leq A_{\text{valid}}. \tag{12}$$

Likewise, invariance cannot identify the division between correct and shared-wrong agreement. These algebraic observations explain why accuracy, R, D, U, coverage, and chance correction should be presented jointly rather than ranked as interchangeable notions of reliability.

4.3 Paired inference, dependence, and multiplicity

The primary cross-language accuracy effect is

$$\widehat{\Delta}_{\mathrm{TR-EN}} = N^{-1} \sum_{i} (C_{i,\mathrm{TR}} - C_{i,\mathrm{EN}}), \tag{13}$$

using identical planned coordinates and counting invalid answers wrong. Effects are evaluated within declared model, size, query-type/distance, relation, presentation, and name-control strata. Any pooled summary must specify whether it weights derived queries, latent orders, or size levels; those are different estimands. Paired McNemar inference (McNemar, 1947) compares discordant correctness outcomes when matched units are independent. The exact conditional version uses a binomial reference on discordant pairs. Repeated queries or presentations from one latent order violate ordinary row-level independence; such rows require cluster-aware uncertainty rather than an unqualified McNemar p-value.

Confidence intervals use 2,000 matched latent-order bootstrap replicates, preserving every dependent query/presentation record and resampling within declared size strata. This extends the resampling principle (Efron, 1979) to the clustered design (Field and Welsh, 2007). Wilson intervals (Wilson, 1927) are useful binomial summaries for R only under an appropriate independent-Bernoulli interpretation; when rows share latent orders, dependence-aware bootstrap intervals carry the inferential interpretation. Intervals are pointwise, not simultaneous bands.

Holm correction (Holm, 1979) applies within predeclared model/design/metric families, with missing planned hypotheses retaining their correction slots. The implementation is not study-wide familywise control across every size, relation, distance, and intervention. Confirmatory families must be specified before outcomes are observed. For base–adapter comparisons, all-planned accuracy, strict R, and coverage are primary; common-four-valid differences in conditional R, invariance, and kappa are explicitly labeled sensitivity analyses and report both arms' valid support.

4.4 Lexical associations and identifiable effects

For each relation direction, the predicted YES/EVET rate is the number of parsed affirmative answers divided by the number of parsed responses in that direction. Its denominator, planned gold-label balance, and invalid rate must be shown together. This descriptive rate may use lone valid responses; it is distinct from the pooled q in Equation (11), which uses jointly valid pairs only. A direction-specific accuracy change and invariance change need not agree.

Frequency–accuracy and tokenization–invariance analyses remain prospective because model outcomes are absent. Spearman association (Spearman, 1904) can summarize monotone relations, but a handful of repeated lexical units does not create independent observations through prompt replication. A multivariable specification may include model, relation family, $\log_2 n$, query distance, and prompt length, with uncertainty clustered by latent order and appropriate attention to lexical reuse. Saturated model-by-vocabulary effects may absorb all between-term frequency variation; unsupported or collinear fits must remain unavailable.

We distinguish a total paired language contrast from an estimate adjusted for token length. Length may lie on a pathway by which language changes the presented input, so adjustment changes the question rather than automatically correcting a nuisance. These are descriptive associational analyses, not identification of a causal frequency or morphology effect. No regression coefficients, significance tests, or model-performance intervals are estimated in the current article.

Table 2: Research questions and the measurements needed to answer them. All model-outcome estimands are pending.

RQ	Matched contrast	Required readout
RQ1	English vs. Turkish at fixed latent query, names, and presentation	Paired accuracy difference, invalid rates, coverage.
RQ2	Turkish canonical vs. equivalent/P1–P5	R, D, U, I, κ, strict R, support.
RQ3	Size and distance within each query type	Separate n and d profiles; endpoint/internal contrast.
RQ4	Relation terms and directions on identical instances	Frequency/token features, predicted YES rate, accuracy and invariance jointly.
RQ5	Direct vs. reasoning; base vs. trained adapter	Matched effects, common support, usage and compute budgets.

5 Experimental Design and Reproducibility

5.1 Research questions and registered contrasts

Table 2 connects each research question to a planned estimand. Four-cell balancing crosses query direction and gold answer. Frozen test strata contain 200 source orders per size and 50 examples per direction–gold cell; smaller pilot/development/training strata use approximately balanced 12/13 cells where 50 cannot be split equally. Derived presentations retain those source labels rather than creating additional independent cases.

The recorded split generation seeds are 401 (training), 402 (development), 403 (pilot), and 404 (test), with presentation seed 431. Seed derivation remains versioned and namespaced. The experiment matrix separates English canonical/equivalent, Turkish canonical/equivalent, cross-language canonical, cross-language equivalent, Turkish ablations, and endpoint-to-interior contrasts. The last changes the queried proposition and may change the gold answer; it is a query-complexity comparison, not an answer-preserving presentation pair. We therefore do not compute presentation invariance by blindly pairing endpoint and internal answers as if they answered the same question.

5.2 Prompts, parsers, and generation budgets

The direct Turkish instruction asks the model to use the supplied premises and return only `EVET` or `HAYIR`; the English control requests `YES` or `NO`. A strict language-specific parser accepts only the specified form, records wrong-language outputs separately, and rejects contradictory or extra answers. It does not search arbitrary response text for a convenient label. Reasoning conditions instead require an unambiguous final-answer marker, with correctness determined by that answer rather than by the explanation.

A run must record immutable model and tokenizer identities, provider/serving revision, sampling parameters, seed where supported, system/user prompts, context and output budgets, reasoning setting, token usage, truncation, and cost status. Model unavailability cannot silently select a replacement. Planned tokens are not billed usage, and isolated-string token counts are not chat-template counts. Context failures, missing completions, and invalid responses stay in planned-sample denominators.

Dry-runs estimate call counts before a small real pilot. Full evaluation requires an approved pilot; large evaluations, including pilots or training evaluation exceeding 2,000 planned generation calls, also require dataset- and scope-bound human linguistic review. Checkpointing and append-only raw records support resumption without overwriting model evidence. An identical-prompt repeat control should accompany stochastic or provider-nondeterministic settings, so cross-presentation disagreement is not automatically attributed to linguistic variation. This is a prospective control, not a reported experiment. A local journal establishes recorded lineage, not independent provider attestation. Current model registries remain disabled and no real model run is reported.

5.3 Reasoning and optional training interventions

Direct answering, open stepwise prompting, structured normalization/direction/transitivity prompting, and provider reasoning-effort controls are distinct arms. Their latent queries, gold labels, and sampling conditions should be held constant where possible. Output budgets must be sufficient and recorded; changes in budget can otherwise confound interpretation. CoT and sampled self-consistency (Wei et al., 2022; Wang et al., 2023) motivate interventions but are not the same as invariance across equivalent presentations. Final-answer improvements do not establish explanation faithfulness (Turpin et al., 2023).

Optional training compares a base model, SFT-LoRA, paired-presentation SFT, SFT with a consistency term, correctness-only GRPO, and correctness-plus-consistency GRPO. LoRA supplies a parameter-efficient adaptation mechanism (Hu et al., 2021); GRPO is a group-relative policy optimization method introduced in mathematical reasoning (Shao et al., 2024). We do not claim to reproduce their original numerical results. The proposed reward form is $r = \text{correctness} + 0.5\,\text{agreement bonus}$, with correctness monitored independently. Under binary rewards, within-group advantage normalization can make an additive agreement component ineffective; diagnostics of reward and normalized advantage are therefore necessary.

Training uses $n \leq 128$, with larger-size and held-out-vocabulary evaluation kept separate. Latent instances must be disjoint across training and test. English context words such as *higher/lower* can overlap even when relation terms are held out, so predicate novelty is not full-context novelty. Fair comparison records example counts, actual training tokens, GPU time, steps, and trainable/update budgets; equal steps do not equalize compute. The local Apple M1/16-GiB preparation machine has no CUDA device. No adapter training, CUDA smoke run, or optional Torch numerical/gradient validation is claimed in this article.

6 Construction and Preprocessing Results

The following results concern actual dataset inventories and retained corpus/tokenizer measurements. They are not simulated model responses. Counts summarize the constructed collection; they are not random-sample estimates of a population accuracy, so model-performance confidence intervals and significance tests are inapplicable. The 33 split-by-size dataset records and the 236 native lexical publication records supply every numerical display in this section.

6.1 Dataset scale and the distinction between units

The collection contains 3,000 latent orders, 23,200 source queries, 5,637,600 prompt identities, and 7,819,200 comparison cells. Source-order allocation is 450 training, 450 development, 300 pilot, and 1,800 test orders. The training allocation spans the frozen size grid, while optional training selects only $n \leq 128$; split membership does not imply selection for optimization. These counts describe different units: each source order supports multiple queries, and each query expands into 243 language/name/relation/presentation identities. Comparison cells additionally include matched contrasts and endpoint-to-interior comparisons. They are not 7.8 million independent reasoning trials.

Figure 2 shows why entity count, independent sample size, and workload must be kept separate. The full-test number of source orders is fixed per n, while the number of feasible distance conditions grows with n. Pilot coverage stops at $n = 128$. The larger prompt workload at high n therefore reflects both presentation length and the design's query expansion; it is not evidence that a model performed more logical steps.

6.2 Independent corpus frequency and morphological realization

The retained lexical measurement uses the three corpus partitions of UD Turkish BOUN release r2.18, revision `0def94a229dcbc28d32192b0138b560c075c6d87`, under CC-BY-SA-4.0. BOUN's resource

Table 3: Verified dataset inventory. Orders are source chains; queries and presentations are dependent derivatives. Presentation counts describe the frozen design, not completed model calls.

Split	Orders	Queries	Presentations	Pairing cells
Training	450	3 550	862 650	1 197 450
Development	450	3 550	862 650	1 197 450
Pilot	300	1 900	461 700	634 500
Frozen test	1 800	14 200	3 450 600	4 789 800
Total	3 000	23 200	5 637 600	7 819 200

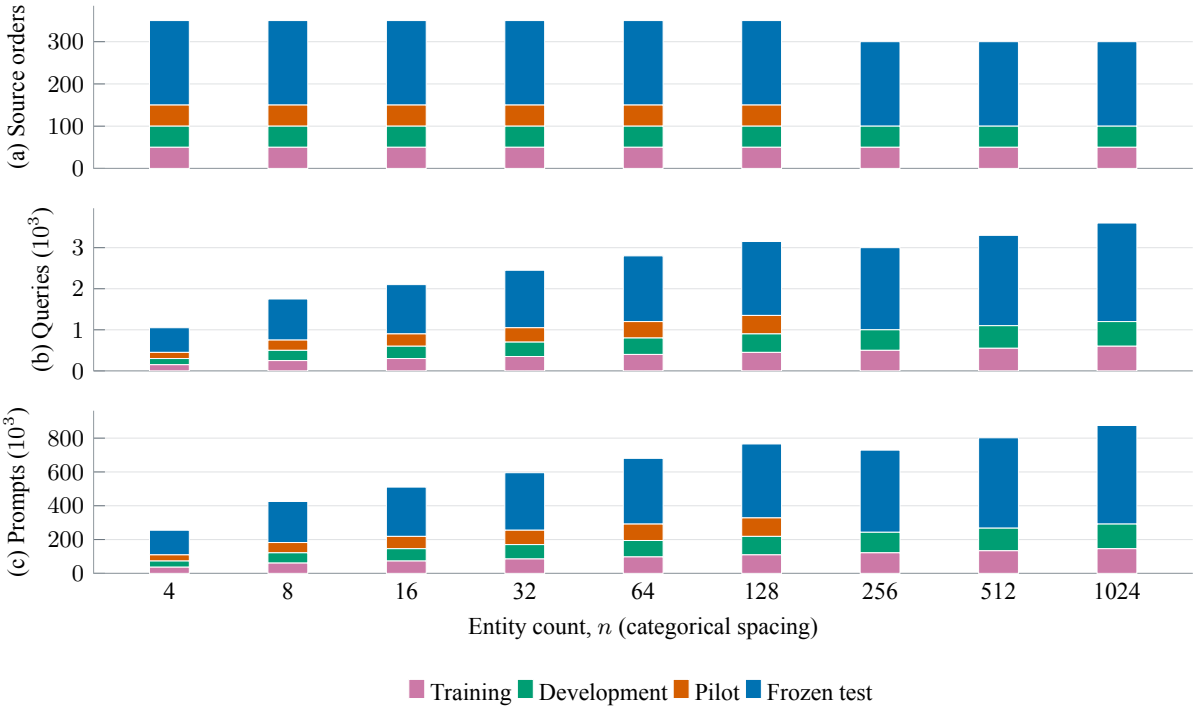


Figure 2: Verified construction counts by size and split. Source orders and derived queries/prompts are different units. This figure contains no model predictions.

design and annotation context are described by Türk et al. (2022); the locked release, not the paper's headline counts alone, defines our measurements. Retained coverage includes 9,761 sentences, 101,052 eligible surface tokens, and 104,424 eligible lemma positions. The recorded normalization uses Unicode NFC and Turkish-aware case folding (I → ı, İ → i), with exact normalized n-gram matching. Surface n-gram opportunity counts differ with expression length; rates must retain their own denominators.

All 18 canonical inflected relation expressions have measured surface count zero, while their recorded lemma counts range from 10 to 390. Figure 3 reports those values and directions. A measured zero is a property of this corpus and matching rule; it is not missing data and does not show that the expression is ungrammatical or absent from all Turkish. Conversely, a positive count of a lemma such as *üst* can include senses outside the intended ordering relation. Surface and lemma frequencies answer different questions.

For every zero surface count, an unsmoothed $\log_{10}$ frequency is undefined. Forward/inverse ratios with zero inverse denominators are likewise undefined; $0/0$ is not set to one. Smoothing could define a different exploratory feature but would require an explicit new measurement specification. We retain the original zero/undefined distinction. These findings motivate careful lexical feature construction rather than a claim that lower frequency causes a reasoning error.

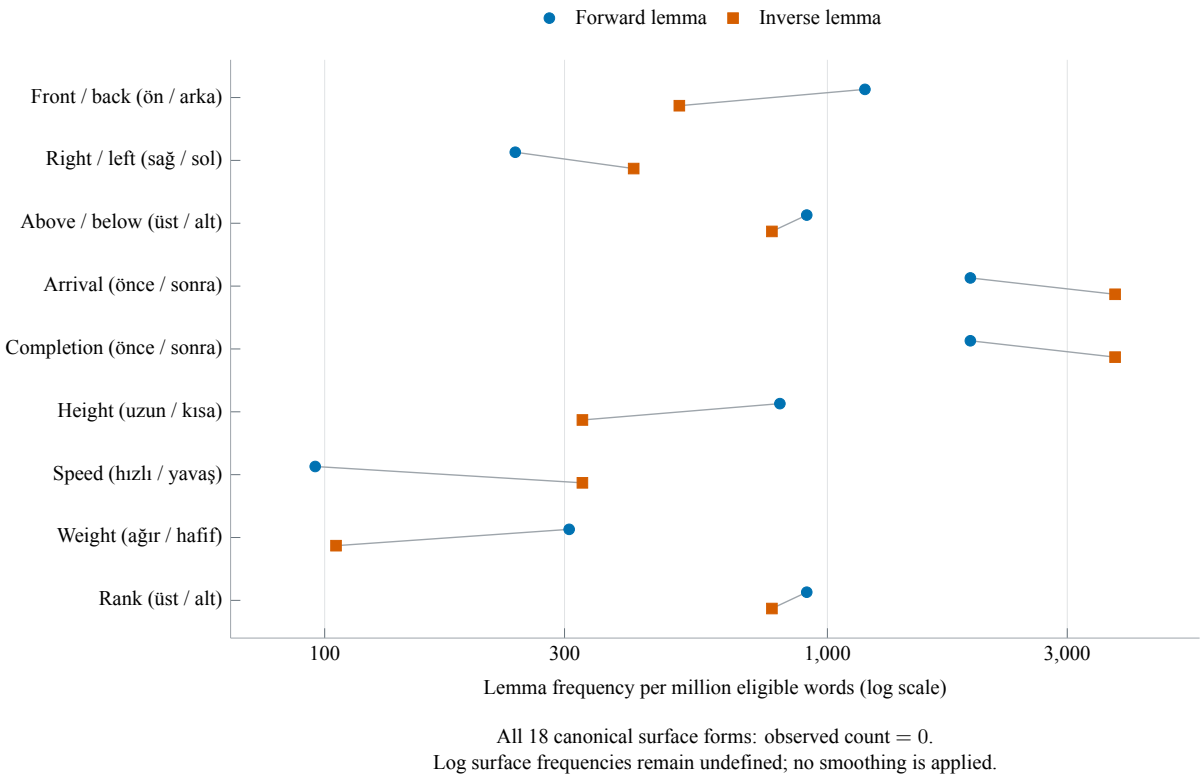


Figure 3: Observed canonical surface counts and lemma frequency proxies for the nine forward/inverse relation pairs. Surface zeros are measured zeros; lemma counts are not sense-disambiguated counts of the full relation expression. Lines connect paired forward and inverse proxies; they are not confidence intervals. No model outcome is represented.

Table 4: Independent relation-lemma counts in UD Turkish BOUN r2.18 (104,424 eligible lemma tokens). These are lemma counts, not sense-disambiguated relation frequencies. All 18 canonical surface-form counts are zero. Shared lemmas recur across templates.

Relation	Forward lemma	Count	Inverse lemma	Count	f_F/f_I
Front / back	ön	124	arka	53	2.34
Right / left	sağ	25	sol	43	0.58
Above / below	üst	95	alt	81	1.17
Arrival	önce	201	sonra	390	0.52
Completion	önce	201	sonra	390	0.52
Height	uzun	84	kısa	34	2.47
Speed	hızlı	10	yavaş	34	0.29
Weight	ağır	32	hafif	11	2.91
Rank	üst	95	alt	81	1.17

6.3 Tokenizer measurements and missing assets

The isolated-vocabulary export has 216 records: six registered configurations, nine relations, two directions, and two languages. Counts are available for the GPT-4o encoding configuration and three Qwen configurations. The two Llama configurations have unavailable local tokenizer assets, so their token lengths remain unmeasured rather than being inferred from another tokenizer. A tokenizer measurement does not mean that model weights were loaded or that its answers were evaluated.

Of the 216 isolated-vocabulary records, 144 are measured and 72 are unavailable. Across the 18 matched relation-term pairs in each measured configuration, the median Turkish/English term-token ratio is 2.75 for the GPT-4o encoding and 2.25 for each Qwen configuration; observed ratios across measured config-

Table 5: Tokenizer preprocessing coverage, not inference results. Each configuration has 18 Turkish and 18 English relation terms; ratios use matched isolated terms. The three Qwen configurations yield identical term counts and are not independent tokenization samples. Missing assets remain unavailable.

Configuration	Measured	Median TR	Median EN	Median TR/EN
GPT-4o / o200k_base	36/36	4.0	2.0	2.75
Qwen2.5-0.5B / HF	36/36	5.0	2.5	2.25
Qwen2.5-1.5B / HF	36/36	5.0	2.5	2.25
Qwen2.5-1.5B / vLLM	36/36	5.0	2.5	2.25
Llama3.2-1B / HF	0/36	—	—	—
Llama3.2-3B / HF	0/36	—	—	—

Table 6: Outcome availability. NA is unavailable, not zero performance. No model-effect estimate or inferential statistic is reported.

Target	Required evidence	Current status
Turkish–English gap	Matched genuine answers and all-planned accuracy	Not run
Morphosyntactic robustness	Canonical/equivalent answers, $R/D/U$, coverage	Not run
Complexity/distance profiles	Answers separated by n, d, and query type	Not run
Lexical bias associations	Direction-specific predictions joined to measured features	Not run
Reasoning/adaptation gains	Matched intervention runs and recorded budgets	Not run

urations range from 0.75 to 6. These are medians of matched ratios, not ratios of median token counts. They summarize the selected lexical inventory without inferential confidence intervals.

Figure 4 compares measured term tokenization across languages and directions on matched lexical entries. Shared-tokenizer configurations are shown as configurations, not treated as independent model families or independent replications. A ratio above one indicates a longer Turkish tokenization for that lexical measurement and tokenizer. It does not imply a corresponding accuracy loss. Isolated expressions, complete plain-text prompts, chat templates, and provider-billed usage have different scopes (Rust et al., 2021; Ahia et al., 2023; Petrov et al., 2023).

The corpus and tokenization audit provides features for RQ4 but cannot answer its performance-association component. The tokenized prompt preview is bounded to $n = 4$; the isolated-term export should not be described as an analysis of full chat length across the entire benchmark. Future outcome analyses must join by the exact latent/presentation coordinate and model/tokenizer revision.

6.4 What remains empirically unanswered

Table 6 prevents preprocessing observations from being mistaken for answers to RQ1–RQ5. In particular, no accuracy-versus-n curve, $R/D/U$ decomposition of model behavior, language-effect interval, or training-gain plot can yet be drawn honestly. The figures in this article instead visualize actual construction/feature measurements and a conceptual pipeline. No source-paper model scores are relabeled as Turkish-CBE results.

7 Discussion

7.1 From translation control to linguistic construct validity

Matching a latent order controls the intended proposition, but it does not ensure that two sentences are equally natural, equally familiar, or equally easy to parse. A Turkish postverbal constituent may preserve truth while changing discourse preference. A morphological alternative may alter word length and lexical realization at the same time. These distinctions connect Turkish-CBE to linguistic evaluation work (Ismay-

Matched isolated-term token count: Turkish / English

	GPT-4o o200k_base	Qwen 0.5B HF	Qwen 1.5B HF	Qwen 1.5B vLLM	Llama 1B missing	Llama 3B missing
Front / back · F	1	1.33	1.33	1.33	—	—
Front / back · I	4	2.5	2.5	2.5	—	—
Right / left · F	1	1.25	1.25	1.25	—	—
Right / left · I	0.75	1	1	1	—	—
Above / below · F	3	4	4	4	—	—
Above / below · I	3	4	4	4	—	—
Arrival · F	4	4	4	4	—	—
Arrival · I	4	4	4	4	—	—
Completion · F	5	6	6	6	—	—
Completion · I	5	6	6	6	—	—
Height · F	1.33	1.67	1.67	1.67	—	—
Height · I	1.67	2	2	2	—	—
Speed · F	1.33	1.33	1.33	1.33	—	—
Speed · I	1.67	1.67	1.67	1.67	—	—
Weight · F	1.33	2	2	2	—	—
Weight · I	2.5	2	2	2	—	—
Rank · F	3	3.5	3.5	3.5	—	—
Rank · I	3	3.5	3.5	3.5	—	—

F: forward; I: inverse. Darker blue indicates a larger token ratio.
Gray cells: missing tokenizer assets, not zero tokens.
Isolated strings only; no model weights or model responses.

Figure 4: Verified isolated relation-term tokenization. Matched Turkish/English measurements and forward/inverse differences describe the lexical strings and retained tokenizer assets. Missing Llama measurements are explicit; these are not model-performance comparisons.

ilzada et al., 2025; Başar et al., 2025; Başar and Bisazza, 2026) without making grammatical acceptability interchangeable with logical accuracy. Human adjudication should therefore evaluate more than whether a parser recovers the expected relation.

The zero exact surface counts illustrate a related measurement issue. An inflected, multiword canonical predicate can be absent under exact matching even when its constituent lemma is frequent. The difference is substantively meaningful for the feature definition, but neither variable directly measures exposure in a proprietary model's training corpus. Lemma ambiguity and corpus domain must remain part of the interpretation. Tokenizer studies likewise motivate explicit measurement (Rust et al., 2021; Ahia et al., 2023; Petrov et al., 2023), not a presumption that subword length is the unique cause of an error.

7.2 Correct consistency versus stable error

The paired decomposition supports questions that a marginal accuracy table cannot answer. A small average difference between canonical and equivalent prompts can coexist with large, offsetting correct-to-wrong and wrong-to-correct flows. Conversely, agreement can improve because a model repeats a wrong label. Equations (9)–(10) specify the evidence needed to distinguish these cases: shared-valid $R/D/U$, all-planned accuracy and strict R, invalid rates, and coverage. Comparisons that condition on different

subsets can otherwise make a selectively valid model appear more robust.

This interpretation also limits the role of chance correction. The pooled-marginal baseline discounts a particular independent-answer bias reference. It is not a causal model of failure, nor does a high coefficient show that the same inferential algorithm was applied across forms. Correctness and invariance remain behavioral properties. Explanation text adds another observation channel but not automatic access to the model's internal computation (Turpin et al., 2023).

7.3 Complexity and intervention claims

The endpoint shortcut makes a strong endpoint score ambiguous as evidence of transitive reasoning. Internal/distance-controlled questions reduce one specific shortcut while retaining a highly structured task family; they do not eliminate every efficient alternative strategy. Changes with n may reflect retrieval, distraction, context length, or the distribution of queried distances. Any future complexity claim must therefore name its query stratum and input/presentation condition.

Reasoning prompting may change token budgets and final-answer formatting as well as intermediate generation. Provider reasoning effort is a different intervention from an explicit verbal instruction. Similarly, SFT or GRPO may strengthen a learned vocabulary mapping without producing broad transfer. Prior work on prompting, aggregation, and adaptation (Wei et al., 2022; Wang et al., 2023; Hu et al., 2021; Shao et al., 2024) supplies methodological options, not evidence for an effect in this benchmark. A finite improvement can extend successful coverage without establishing competence throughout the family.

7.4 Status of the scientific claims

The supported conclusions are restricted to the constructed design and recorded preprocessing measurements. The framework makes comparison units, linguistic transformations, missingness, and inferential assumptions explicit. It also exposes gaps that must be filled: human review, actual model calls, declared primary test families, and appropriate training hardware if optional adaptation is attempted. Neither an expected directional hypothesis nor a favorable narrative is substituted for missing outcomes. This separation allows future empirical findings, including null or adverse effects, to be incorporated without changing the underlying definitions after observing results.

8 Limitations

Empirical availability. No verified model inference or adapter training outcome is available. Dataset counts and corpus/tokenizer measurements are real, but they do not support claims about model correctness, language gaps, or intervention efficacy. The draft is not a completed empirical evaluation or an externally preregistered study.

Linguistic coverage. The renderer covers a closed grammar and a finite relation inventory. Shared generation/parsing code can conceal a common mistake. Human naturalness and equivalence review is pending. Turkish morphology and word order are richer than the implemented templates, and P4/P5 do not identify a pure morphology manipulation. Minimal-pair results in prior literature cannot validate our stimuli automatically.

Task and sample scope. Strict total orders exclude partial orders, contradictory premises, ties, and open-world uncertainty. Renamed orders are structurally isomorphic. Multiple queries, names, relations, and presentations share each latent source; millions of records do not supply millions of independent samples. Exact latent split separation does not establish absence of model pretraining contamination.

Frequency and tokenization proxies. BOUN has finite genre/sense coverage. Surface n-gram and lemma opportunities differ, exact canonical zeros need not reflect general language usage, and lemma counts are not relation-sense counts. Missing tokenizer assets remain missing. Isolated-string measurements exclude chat overhead and cannot establish input truncation or billed usage. Correlations with model performance cannot be estimated without actual predictions.

Inference and missingness. Context overflow, wrong-language answers, truncation, and provider nondeterminism may be systematically associated with difficult items. Conditional CBE statistics require coverage and all-planned counterparts. Cluster bootstrap intervals depend on sufficient independent latent orders; Wilson intervals alone do not account for repeated derived rows. The declared Holm families do not provide study-wide simultaneous error control.

Training feasibility and transfer. The recorded M1/16-GiB environment is not a CUDA training environment. Optional Torch numerical/gradient tests and real training smoke validation remain unexecuted; direct dependency pins are not a fully resolved transitive CUDA lock. Equal steps do not equalize examples, tokens, GPU time, or update budgets. Held-out relation terms can share context vocabulary, limiting lexical-transfer interpretations.

9 Conclusion

Turkish-CBE provides a controlled framework for studying the relationship between logical correctness and presentation stability in Turkish and English. Its contribution is the explicit connection between latent ordering, Turkish linguistic realization, query geometry, name controls, and denominator-aware paired analysis. The constructed collection contains 3,000 independent source-order identities before query and presentation expansion; those identities, rather than derived prompt counts, define the clustering structure.

The available construction and preprocessing results demonstrate the need to distinguish measured surface zeros from lemma familiarity, available from missing tokenizer assets, and raw agreement from correct consistency. They do not establish any model-performance effect. Genuine paired evaluation after human stimulus review is necessary to answer RQ1–RQ5. Future reports should preserve the present definitions, expose uncertainty and missingness, and interpret improvements as evidence about the tested coverage rather than proof of competence over an entire problem family.

References

O. Ahia, S. Kumar, H. Gonen, J. Kasai, D. R. Mortensen, N. A. Smith, and Y. Tsvetkov. Do all languages cost the same? tokenization in the era of commercial language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9904–9923. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.614. URL https://aclanthology.org/2023.emnlp-main.614/.

E. Başar and A. Bisazza. A morphology-aware evaluation of Turkish syntax in large language models. In *Proceedings of the Second Workshop Natural Language Processing for Turkic Languages (SIGTURK 2026)*, pages 95–102. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.sigturk-1.9. URL https://aclanthology.org/2026.sigturk-1.9/.

E. Başar, F. Padovani, J. Jumelet, and A. Bisazza. TurBLiMP: A Turkish benchmark of linguistic minimal pairs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 16495–16510. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.834. URL https://aclanthology.org/2025.emnlp-main.834/.

P. Clark, O. Tafjord, and K. Richardson. Transformers as soft reasoners over language. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20)*, pages 3882–3890. International Joint Conferences on Artificial Intelligence Organization, 2020. doi: 10.24963/ijcai.2020/537. URL https://ijcai.org/proceedings/2020/537.

J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. doi: 10.1177/001316446002000104. URL https://journals.sagepub.com/doi/abs/10.1177/001316446002000104.

B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552. URL https://doi.org/10.1214/aos/1176344552.

Y. Elazar, N. Kassner, S. Ravfogel, A. Ravichander, E. Hovy, H. Schütze, and Y. Goldberg. Measuring and improving consistency in pretrained language models. *Transactions of the Association for Computational Linguistics*, 9:1012–1031, 2021. doi: 10.1162/tacl_a_00410. URL https://aclanthology.org/2021.tacl-1.60/.

C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x. URL https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9868.2007.00593.x.

S. Han, H. Schoelkopf, Y. Zhao, Z. Qi, M. Riddell, W. Zhou, J. Coady, D. Peng, Y. Qiao, L. Benson, L. Sun, A. Wardle-Solano, H. Szabó, E. Zubova, M. Burtell, J. Fan, Y. Liu, B. Wong, M. Sailor, A. Ni, L. Nan, J. Kasai, T. Yu, R. Zhang, A. Fabbri, W. M. Kryscinski, S. Yavuz, Y. Liu, X. V. Lin, S. Joty, Y. Zhou, C. Xiong, R. Ying, A. Cohan, and D. Radev. FOLIO: Natural language reasoning with first-order logic. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 22017–22031. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.1229. URL https://aclanthology.org/2024.emnlp-main.1229/.

S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685v2*, 2021. URL https://arxiv.org/abs/2106.09685.

M. Ismayilzada, D. Circi, J. Sälevä, H. Sirin, A. Köksal, B. Dhingra, A. Bosselut, D. Ataman, and L. van der Plas. Evaluating morphological compositional generalization in large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1270–1305. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.naacl-long.59. URL https://aclanthology.org/2025.naacl-long.59/.

J. Jiang and C. Zhai. Competence-based evaluation of logical reasoning in large language models. *arXiv preprint arXiv:2610.22894v1*, 2026. URL https://arxiv.org/html/2610.22894v1.

J. Liu, L. Cui, H. Liu, D. Huang, Y. Wang, and Y. Zhang. LogiQA: A challenge dataset for machine reading comprehension with logical reasoning. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20)*, pages 3622–3628. International Joint Conferences on Artificial Intelligence Organization, 2020. doi: 10.24963/ijcai.2020/501. URL https://ijcai.org/proceedings/2020/501.

Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. doi: 10.1007/BF02295996. URL https://doi.org/10.1007/BF02295996.

S. I. Mirzadeh, K. Alizadeh, H. Shahrokhi, O. Tuzel, S. Bengio, and M. Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=AjXkRZIvjB.

K. Oflazer. Two-level description of Turkish morphology. *Literary and Linguistic Computing*, 9(2):137–148, 1994. doi: 10.1093/llc/9.2.137. URL https://academic.oup.com/dsh/article-abstract/9/2/137/942916.

A. Petrov, E. La Malfa, P. H. S. Torr, and A. Bibi. Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-1608. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/74bb24dca8334adce292883b4b651eda-Abstract-Conference.html.

M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.442. URL https://aclanthology.org/2020.acl-main.442/.

P. Rust, J. Pfeiffer, I. Vulić, S. Ruder, and I. Gurevych. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3118–3135. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.acl-long.243. URL https://aclanthology.org/2021.acl-long.243/.

A. Saparov and H. He. Language models are greedy reasoners: A systematic formal analysis of chain-of-thought. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.01240.

S. Segura, G. Fraser, A. B. Sánchez, and A. Ruiz-Cortés. A survey on metamorphic testing. *IEEE Transactions on Software Engineering*, 42(9):805–824, 2016. doi: 10.1109/TSE.2016.2532875. URL https://www.isa.us.es/publications/type/article-journal/2016/survey-metamorphic-testing.

Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300v3*, 2024. URL https://arxiv.org/abs/2402.03300.

F. Shi, M. Suzgun, M. Freitag, X. Wang, S. Srivats, S. Vosoughi, H. W. Chung, Y. Tay, S. Ruder, D. Zhou, D. Das, and J. Wei. Language models are multilingual chain-of-thought reasoners. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=fR3wGCk-IXp.

K. Sinha, S. Sodhani, J. Dong, J. Pineau, and W. L. Hamilton. CLUTRR: A diagnostic benchmark for inductive reasoning from text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4506–4515. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1458. URL https://aclanthology.org/D19-1458/.

C. Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904. doi: 10.2307/1412159. URL https://www.jstor.org/stable/1412159.

O. Tafjord, B. Dalvi, and P. Clark. Proofwriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP*

2021, pages 3621–3634. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.findings-acl.317. URL https://aclanthology.org/2021.findings-acl.317/.

U. Türk, F. Atmaca, Ş. B. Özateş, G. Berk, S. T. Bedir, A. Köksal, B. Ö. Başaran, T. Güngör, and A. Özgür. Resources for Turkish dependency parsing: introducing the BOUN treebank and the BoAT annotation tool. *Language Resources and Evaluation*, 56(1):259–307, 2022. doi: 10.1007/s10579-021-09558-0. URL https://link.springer.com/article/10.1007/s10579-021-09558-0.

M. Turpin, J. Michael, E. Perez, and S. R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-3275. URL https://papers.neurips.cc/paper_files/paper/2023/hash/ed3fea9033a80fea1376299fa7863f4a-Abstract-Conference.html.

X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2203.11171.

J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022. doi: 10.52202/068431-1800. URL https://proceedings.neurips.cc/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html.

E. B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953. URL https://doi.org/10.1080/01621459.1927.10502953.

W. Yu, Z. Jiang, Y. Dong, and J. Feng. ReClor: A reading comprehension dataset requiring logical reasoning. In *The Eighth International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/2002.04326.

A Elementary Identities and Chance References

Proposition 1. *For a binary task with a common gold answer and two jointly valid answers, the three outcome masses partition the sample, and* $A_{\text{valid}} = R + U/2$, $I = R + D$.

Proof. Each pair is both correct, both wrong, or exactly one correct, so $R + D + U = 1$. The two-presentation correctness count is two for a both-correct pair, zero for a both-wrong pair, and one for a discordant pair. Dividing by two yields $A_{\text{valid}} = R + U/2$. In a binary task, both-wrong answers are the same wrong label; agreement is therefore $R + D$. The argument requires the same valid-pair support and the same gold proposition. □

Proposition 2. *For* $N > 0$ *and* $M > 0$*, the strict robustness identity is* $R_{\text{strict}} = vR$*, and the accuracy-constrained bounds are* $\max(0, 2A_{\text{valid}} - 1) \leq R \leq A_{\text{valid}}$.

Proof. Both robustness numerators count the same both-correct valid pairs; their denominators are M and N. Multiplying by M/N gives the first identity. Since $U = 2(A_{\text{valid}} - R) \geq 0$, $R \leq A_{\text{valid}}$. Since $D = 1 - R - U = 1 - 2A_{\text{valid}} + R \geq 0$, $R \geq 2A_{\text{valid}} - 1$; probabilities are also nonnegative. □

Proposition 3. *Let* p_0, p_1 *be the two YES marginals on jointly valid pairs and* $q = (p_0 + p_1)/2$*. The CBE pooled reference and Cohen's separate-marginal reference satisfy*

$$I_{e,\text{CBE}} - I_{e,\text{Cohen}} = \frac{(p_0 - p_1)^2}{2}, \tag{14}$$

where $I_{e,\text{Cohen}} = p_0 p_1 + (1 - p_0)(1 - p_1)$.

Proof. Expanding the pooled expression gives $1 - p_0 - p_1 + (p_0 + p_1)^2/2$. Subtracting $1 - p_0 - p_1 + 2p_0 p_1$ yields the stated square. Thus the references agree when the marginals agree; otherwise they are distinct. For fixed observed agreement below one, increasing the chance reference decreases the coefficient wherever its denominator is positive. This is an algebraic property, not an empirical model result or a new estimate of reliability. □

B Illustrative Turkish Equivalence and Its Boundary

The following forms illustrate the closed-grammar contract; they are not model outputs:

TR forward: Ayşe, Bora'nın önündedir.
TR converse: Bora, Ayşe'nin arkasındadır.
EN forward: Ayşe is in front of Bora.
EN converse: Bora is behind Ayşe.

The equivalence presupposes the specified linear interpretation and the same entities. Proper-name suffixes use the pronunciation and vowel/consonant environment of the supplied name; ASCII transliteration is a separate name control, not an alternative spelling silently substituted in a Turkish condition.

For distinct a, b in a strict total order, trichotomy gives $\neg(a \prec b) \Leftrightarrow b \prec a$. This justifies a formal negation transformation only under those assumptions. It does not license unrestricted natural-language negation, ties, or partial-order interpretations. Optional P6 stimuli require their own grammar and human review and are excluded from the frozen core described in the main text.

C Data, Figure, and Bibliography Provenance

Every new quantitative figure reads the retained dataset-statistics or lexical-publication CSVs. The plot manifest records source SHA-256 hashes, deterministic derived values, and generated asset hashes. Surface count zero is retained as zero; undefined logarithms/ratios and unavailable tokenizer assets remain unavailable. The retained plain-text exports are not reclassified as provider-billed usage. Figure 1 is a conceptual diagram with an illustrative instance; Figures 2–4 show construction/preprocessing measurements only.

The companion bibliography records primary publication URLs/DOIs checked on 26 October 2026. Related-work claims are bounded to the corresponding task and language scope. In particular, MGSM is not a Turkish benchmark, ParaRel is not a logical-ordering benchmark, and grammatical minimal-pair evaluations are not answer-preserving inference pairs. The complete original CSV and compact table exports remain available with their native stratum identifiers; the paper uses selected summaries for readability rather than treating all dependent rows as new evidence.

C.1 Rebuilding the manuscript

The entry point is `paper/main.tex`; chapter-like units are plain `.tex` files under `paper/sections/`. Compilation uses XeLaTeX and BibTeX with a Unicode-capable font and requires no model API, model weights, or experiment execution. The prior Markdown snapshot and its existing exported values are preserved for provenance; this LaTeX manuscript is the new reading and editing entry point.